

9. Linux

Linux is licensed under the GNU GPL

it is GNU Linux, they became one.

they are open source meaning anyone can read, audit, use, modify or contribute, its also free.

GPL -> General Purpose License

GNU is the top part, Linux is just the kernel

one linux kernel -> thousands of distros

softwares licensed under GPL are open source and can be commercialized but the source code should be readily available and open

Distros are made up of:

kernel, packages, package manager, desktop UI.

UBUNTU 24.04.3 LTS. LTS -> Long Term Support

download and install, 6 cores, 8gb ram, then take a snapshot after setting up the os.

make it full screen, make it dark mode.

install the cd guest additions and enable the bidirectional clipboard.

man sudo_root -> manual for sudo

~\$ means we are in the home directory of the standard local user.

birat@birat-ubuntu-phd -> birat is the username and after @ is the computer name.

pwd -> present working directory

ls -> display the contents of the directory

cd Documents -> changes the directory to Documents folder.

cd -> changes the directory

cd .. -> goes back one directory in the path.

Every thing in Linux is case sensitive.

pressing TAB will autocomplete the name of the directory.

`mkdir` -> makes a directory

`touch` -> creates a file

`gnome-text-editor test.txt` -> opens the test.txt

`cat test.txt` -> shows the contents of the test.txt

`cat --help` -> opens up the help page for cat

`man cat` -> gives more details for cat

`cat /etc/passwd` -> see full list of users

root user is the administrative user.

password is in the hashes of the shadow file in etc folder which only the root user has access too. previously the x file used to have it, but it was not secure as all the users had access to it.

`whoami` -> shows which user is using the system. sometimes using remote machines, you would not know who you are.

`sudo` command lets you do what root user is allowed to do.

`sudo -l` lists what commands the current user can run using the `sudo` command.

in the /etc folder, `sudo cat shadow` shows all the hashes of the passwords.

`su root` -> asks for password for root account to change the user to root user.

`sudo su root` -> also does the same thing.

root@birat-ubuntu-phd:/etc# =>>says that we are running as root.

`ctrl + C` will exit the prompt.

`cd /` -> will take you to the root of the filesystem

filesystem hierarchy system says what files and folders should be in the root in linux for it to operate.

/bin -> has the binaries, they can just run without putting the whole path in the prompt because these are in the path, so by default, this folder is searched for the binary's command.

/sbin -> this folder can only be used by the root user. it has commands like `deluser` which deletes the user, this command should only be runnable by the root user.

/lib -> this folder has the libraries that the binaries from /bin and /sbin folder require to run.

/usr -> has /bin and /sbin folders which are not essential for the operating system to use but has folders which are for the end user.

etc = Editable Text Configuration

/etc -> this folder has a lot of .conf files which are the configuration files which are for the binaries and also some for the operating systems.

/home -> this directory has all the users created by us as the folders
so, from the root folder:
cd /home/birat would take me to the user birat.

```
birat@birat-ubuntu-phd:/$  
( becomes )  
birat@birat-ubuntu-phd:~$
```

/dev -> this folder has files that are created while connecting any usb device.

/boot -> this has files that are required for the os to boot up.

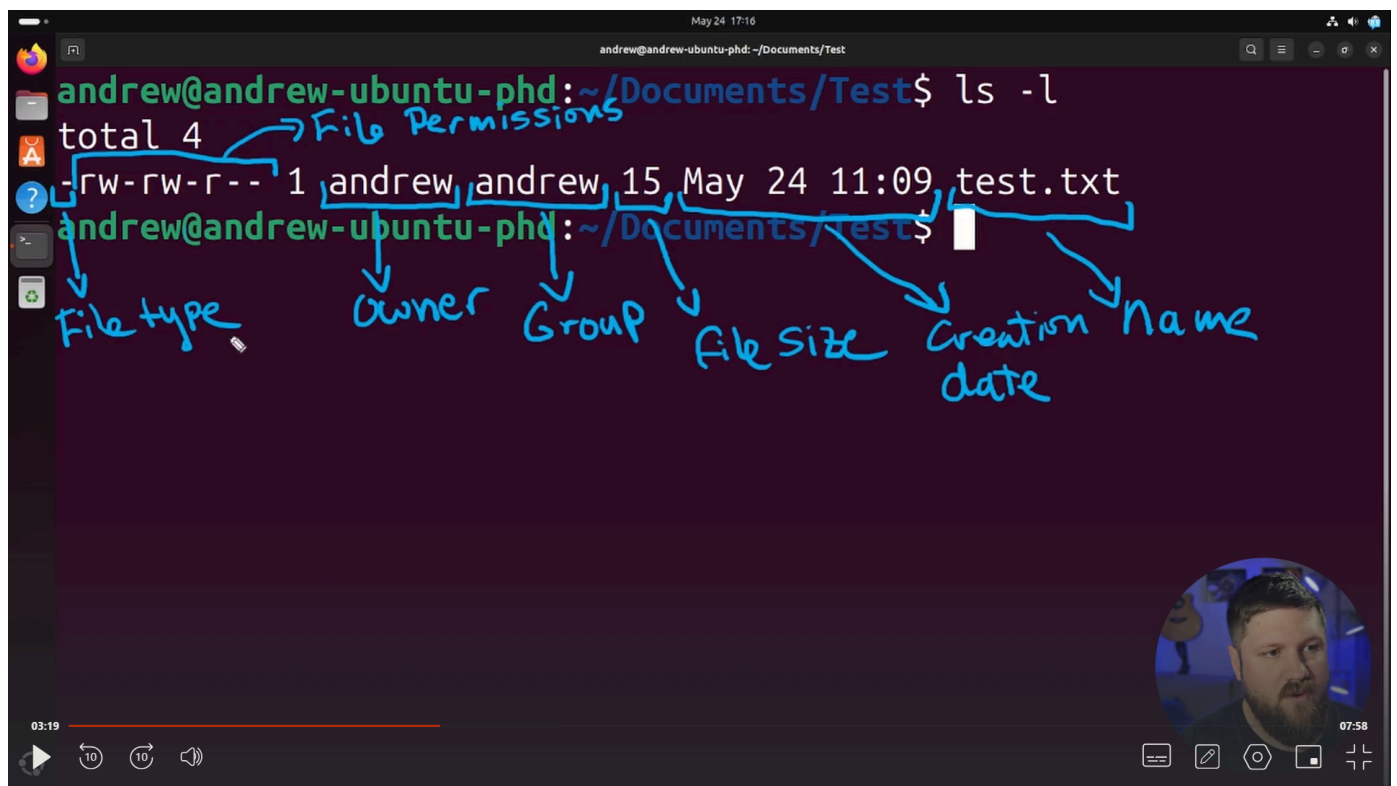
/opt -> has nothing but is designed to keep files for the optional softwares we decide to install.

/tmp -> root or the standard user can write temporary files on this folder. the files on temp folder do not persist between reboots.
after reboot, the files in this folder are deleted.

/var -> stands for variables, os saves variable on the folders and files on this directory.

for example crash statistics are stored for some time in the crash folder.

/proc -> are the processes that are running, these are written on the memory by the os and not in the disks.



```
-rw-rw-r-- 1 birat birat 14 Dec 31 15:57 test.txt
```

(-) -> file type

(rw-rw-r--) -> file permissions

(birat) -> owner

(birat) -> group

(14) -> file size

(Dec 31 15:57) -> creation date

(test.txt) -> file name

first (rw-) -> permissions for the owner

second (rw-) -> permissions for the group

third (r--) -> permissions for everyone else

important to know that this defines security and access because all users can use these permissions.

three permissions which can be turned on or off.

r -> read -> 4 as the binary value

w -> write -> 2 as the binary value

x -> executable -> 1 as the binary value

so for the above example its 664.

owner has 6 (r+w),

group has 6 (r+w),

everyone else has 4 (r).

```
andrew@andrew-ubuntu-phd: ~/Documents/Test$ ls -l
total 4
-rw-rw-r-- 1 andrew andrew 15 May 24 11:09 test.txt
```

File Permissions

File type: - (regular file)
Permissions: rw-rw-r-- (owner: read/write, group: read/write, others: read)
Owner: andrew
Group: andrew
File size: 15
Creation date: May 24 11:09
Name: test.txt

OWNER: andrew
Group: andrew
Everyone: (no permissions)

4 2 1 → 664

chmod -> changes the file permissions.

so, ex: chmod 660 test.txt changes the everyone's else access to none, before it was just read.

```
andrew@andrew-ubuntu-phd: ~/Documents/Test$ ls -l
total 4
-rw-rw-r-- 1 andrew andrew 15 May 24 11:09 test.txt
andrew@andrew-ubuntu-phd: ~/Documents/Test$ chmod 660 test.txt
andrew@andrew-ubuntu-phd: ~/Documents/Test$ ls -l
total 4
-rw-rw---- 1 andrew andrew 15 May 24 11:09 test.txt
```

chown -> changes the owner

chown root test.txt -> changes the owner of the file.

sudo chown root test.txt -> works when you are not root.

```
andrew@andrew-ubuntu-phd: ~/Documents/Test
andrew@andrew-ubuntu-phd:~/Documents/Test$ ls -l
total 4
-rw-rw-r-- 1 andrew andrew 15 May 24 11:09 test.txt
andrew@andrew-ubuntu-phd:~/Documents/Test$ chmod 660 test.txt
andrew@andrew-ubuntu-phd:~/Documents/Test$ ls -l
total 4
-rw-rw---- 1 andrew andrew 15 May 24 11:09 test.txt
andrew@andrew-ubuntu-phd:~/Documents/Test$ chown root test.txt
chown: changing ownership of 'test.txt': Operation not permitted
andrew@andrew-ubuntu-phd:~/Documents/Test$ sudo chown root test.txt
[sudo] password for andrew:
andrew@andrew-ubuntu-phd:~/Documents/Test$ ls -l
total 4
-rw-rw---- 1 root andrew 15 May 24 11:09 test.txt
andrew@andrew-ubuntu-phd:~/Documents/Test$
```

chgrp -> changes the group for the file.

```
andrew@andrew-ubuntu-phd: ~/Documents/Test
andrew@andrew-ubuntu-phd:~/Documents/Test$ sudo chgrp root test.txt
andrew@andrew-ubuntu-phd:~/Documents/Test$ ls -l
total 4
-rw-rw---- 1 root root 15 May 24 11:09 test.txt
andrew@andrew-ubuntu-phd:~/Documents/Test$
```

sudo chgrp root test.txt -> will change group to root.

changing back the owner and the group to previous user and setting rw, r and r permissions for the owner, group and everyone respectively for the file test.txt:

```
andrew@andrew-ubuntu-phd:~/Documents/Test$ cat test.txt
cat: test.txt: Permission denied
andrew@andrew-ubuntu-phd:~/Documents/Test$ ls -l
total 4
-rw-rw---- 1 root root 15 May 24 11:09 test.txt
andrew@andrew-ubuntu-phd:~/Documents/Test$ sudo chown andrew test.txt
andrew@andrew-ubuntu-phd:~/Documents/Test$ sudo chgrp andrew test.txt
andrew@andrew-ubuntu-phd:~/Documents/Test$ ls -l
total 4
-rw-rw---- 1 andrew andrew 15 May 24 11:09 test.txt
andrew@andrew-ubuntu-phd:~/Documents/Test$ chmod 644 test.txt
```



```
andrew@andrew-ubuntu-phd:~/Documents/Test$ ls -l
total 4
-rw-r--r-- 1 andrew andrew 15 May 24 11:09 test.txt
andrew@andrew-ubuntu-phd:~/Documents/Test$
```

on the pic below, d is stands for directory and directories are executable.

```
andrew@andrew-ubuntu-phd:~/Documents/Test$ ls -l
total 4
-rw-r--r-- 1 andrew andrew 15 May 24 11:09 test.txt
andrew@andrew-ubuntu-phd:~/Documents/Test$ cd ..
andrew@andrew-ubuntu-phd:~/Documents$ ls -l
total 4
drwxrwxr-x 2 andrew andrew 4096 May 24 11:09 Test
andrew@andrew-ubuntu-phd:~/Documents$
```

Ticket Interrupt:

add two users to linux host, Bertram and Erlich

create the one time login password and make them create new password. create a new folder in the root directory called "Confidential", make root the owner of that folder and give all the permissions. Allow those two users to have the same level of permissions. everyone else should have no permissions, so they can't access the folder.

sudo useradd ____ -> adds new user

sudo passwd _____ -> creates a new password

sudo useradd -m -s /bin/bash _____

-> creates new user and creates a directory in the home folder, and gives a login shell.

sudo passwd -e ____ -> will expire the password and the user has to change immediately.

sudo groupadd ____ -> creates a group

sudo usermod -a -G (groupname) (username) -> adds the user to the group.

getent group (groupname) ->

lists the members of the group.

sudo apt install members -> installs members to list members in the group.

members (groupname) ->

will list the members of that group.

sudo gpasswd -d (username) (groupname) -> to delete the user from that group.

```
birat@birat-ubuntu-phd:/$ sudo gpasswd -d root GroupForConfidential
Removing user root from group GroupForConfidential
birat@birat-ubuntu-phd:/$
```

sudo deluser (username) (groupname)

-> does the same thing.

sudo usermod -G group1,group2 username -> also does the same thing.

sudo snap install code -> installs vs code for only that user.

sudo apt update -> updates the package index

because of security issues, just typing code won't work.

run vscode as root user -> sudo code --no-sandbox --user-data-dir=/tmp/vscode-root

which whoami -> shows where is the underlying binary for the whoami is.

```
root@birat-ubuntu-phd:/opt# ./backup.sh
Current date read as: 2026-01-03
root@birat-ubuntu-phd:/opt#
```

echo "hello backup" > hello.txt -> create hello.txt and put that line on it.

```
$ backup.sh X
opt > $ backup.sh
1  #!/bin/bash
2
3  # Script to backup the contents of "/Confidential" to a "/Backups/<date>" where date is the current date of the backup
4
5  # Storing the current date
6
7  date=$(date +%F)
8
9  echo "Current date read as: $date"
10 |
11
12 if [ ! -d "/Backups/$date" ]
13 then
14     echo "/Backups/$date doesn't exists, creating directory"
15     mkdir "/Backups/$date"
16 else
17     echo "Warning: /Backups/$date already exists, backup will overwrite"
18 fi
19
20 # # Making directory for the current date inside "/Backups"
21 # mkdir "/Backups/$date"
22
23
24 # Copying the Confidential directory to the backup directory
25 cp -r /Confidential/* /Backups/$date/
26
```



```
root@birat-ubuntu-phd:/Backups/2026-01-03# ls
hello.txt  passwords.txt
root@birat-ubuntu-phd:/Backups/2026-01-03# cat hello.txt
Hi overwrite
root@birat-ubuntu-phd:/Backups/2026-01-03#
```

birat@birat-ubuntu-phd:/home/Bertram\$ cat /var/log/syslog

-> this shows the cronjob log for that user

crontab -l -> lists the cronjobs for that user

crontab -e -> opens the editor (nano)

sudo crontab -u root -e -> opens up the cronjobs script for that user

sudo crontab -u root -l -> lists the cronjobs

package managers are used to keep the packages for installation.

ubuntu is debian based. so it uses apt = Advanced Package Tool

sudo apt update -> updates the packages with the externally available

sudo apt upgrade -y -> says yes to the upgrades available

sudo apt update && sudo apt upgrade -y -> are usually ran together

sudo apt install gedit -> installs the package named gedit

sudo apt remove gedit -> uninstalls the package named gedit

run the system update from the app updater